

Demonstrating Thermotion: Design and Fabrication of Thermofluidic Composites for Animation Effects on Object Surfaces

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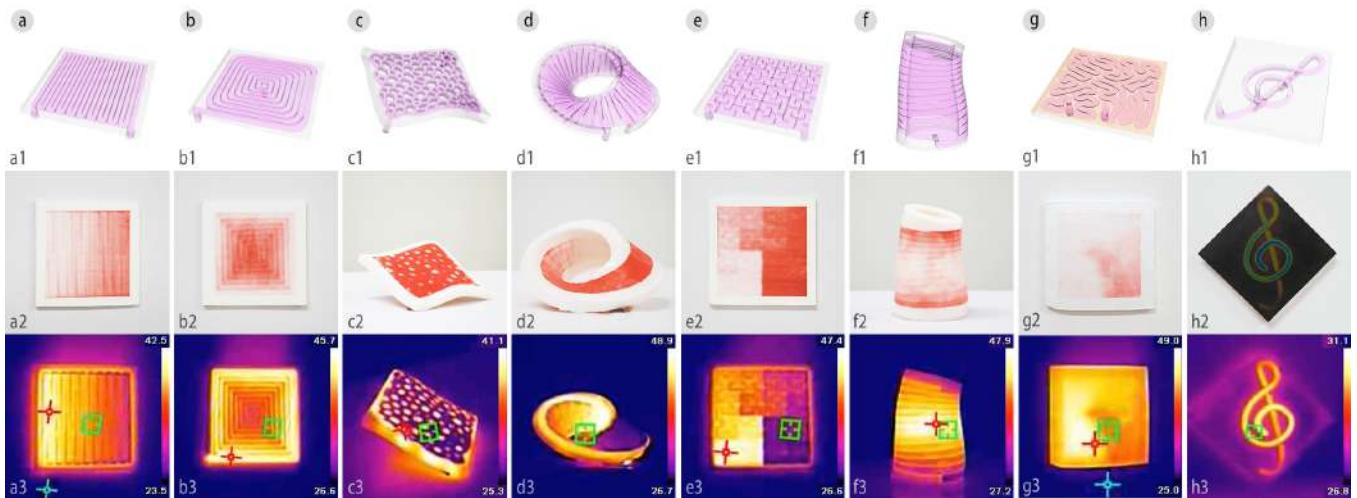


Figure 1: Eight examples of animation effects on two and three dimensional object surfaces by Thermotion, each with captures of the 3D model (fluidic channels in pink and heat diffusion layer in orange), the photography image, and the thermography image captured at the same moment as the photography image. (a-e) Five primitive types of (a) wipe, (b) spread, (c) texture, (d) path, and (e) customization. (f) A spaced version of primitive wipe. (g) Blur effect with a heat diffusion layer. (h) Multi-color effect using liquid crystal for the painting layer.

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ABSTRACT

We demonstrate *Thermotion*, a novel method using thermofluidic composites to design and display thermochromic animation effects on object surfaces. With fluidic channels embedded under the object surfaces, the composites utilize thermofluidic flows to dynamically control the surface temperature as an actuator for thermochromic paints, which enables researchers and designers for the first time to create animations not only on two and three-dimensional surfaces but also on the surface made of a few flexible everyday materials. We demonstrate the fabrication workflow of the composites and the design space of the animation effects. A range of applications

is shown leveraging the objects' dynamic displays both visually and thermally, including dynamic artifacts, teaching aids, and ambient displays. We envision an opportunity to extend thermofluidic composites to other heat-related practices for further dynamic and programmable interactions with temperature.

CCS CONCEPTS

• **Human-centered computing** → **Interactive systems and tools.**

KEYWORDS

Color-changing Interface, Thermofluidic Composites, Thermal Design, Programmable Materials, Computational Fabrication, Human-Material Interaction

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1 INTRODUCTION

Color changing on object surfaces is widely researched in HCI fields. Designers and researchers have explored new interfaces via dynamic and programmable changes in the appearance of physical objects to convey information and aesthetic expression[10, 14, 16]. Recently, an emerging trend aims to achieve dynamic displays in arbitrary shapes (e.g., curved surfaces[1, 4, 6] or flexible surfaces[7, 10, 11]) to expand the functionality and expressiveness of the interfaces, as well as embed display capability in everyday materials (e.g., fabric[2, 3] or paper[5, 9, 15]) to conserve the familiarity and aesthetics of the interfaces for human.

For this purpose, there are various studies on color-changing mechanisms and techniques. A majority of works used thermochromic materials to create color-changing effects due to their applicability to different substrate materials[3]. Researchers have used electronic components including resistive heaters[13] and Peltier elements[12] to activate the thermochromic paints. However, these methods could only support a uniform color-changing effect in one area driven by one electronic component, which greatly increases the volume and complexity of the circuitry to display dynamic and delicate information such as animation effects. Researchers also used photochromic materials to re-program the multi-color appearance of a 3D object[6, 17]. However, this method requires configuring the object in precise optical equipment for exposure for at least one minute, which makes it challenging to utilize the color-changing process as a dynamic display. Other color-changing interfaces rely on fluidic materials[8, 10] or luminous materials[4, 14] to create 3D dynamic displays, while the requirement of transparent or luminous surface limits the surface apart from everyday materials. Overall, few previous works demonstrate the design and fabrication workflow and the necessary technical implementation of a color-changing interface with the capability of dynamic display on complex shapes and compatibility with everyday materials.

To address the above limitations, we introduce *Thermotion*, a novel method using thermofluidic composites with thermochromic paints to create color-changing interfaces, which enables dynamic and programmable animation effects to display on two and three-dimensional object surfaces compatible with flexible everyday materials, e.g., paper and fabric. Specifically, Thermotion allows us to use thermofluidic flows in the channels embedded under the object surfaces to dynamically control the temperature and the color-changing effects of the thermochromic paints on the object surfaces. In this paper, we demonstrated the mechanism and structure of the composites and the fabrication pipeline. We also demonstrated the design space of the animation effects derived from the variation of the structures and the application scenarios of dynamic artifacts, teaching aids, and ambient displays. We expected to use Thermotion to envision an opportunity to extend thermofluidic composites to heat-related practices in HCI (e.g., thermal haptic interfaces or thermoreactive interfaces) to create further dynamic and programmable interactions with temperature.

2 THERMOTION COMPOSITES

2.1 Mechanism and Structure

The fundamental working mechanism of Thermotion is as follows: Thermofluids of different temperatures flow through the channels embedded under the object's surface. As a result of heat conduction, the temperature of the object surface is dynamically changed and controlled. This activates the color-changing effects of the thermochromic paintings on the object's surface, thereby creating animation effects. To systematically utilize the above mechanism, we divided the Thermotion thermofluidic composites into four layers, shown in Figure 2.

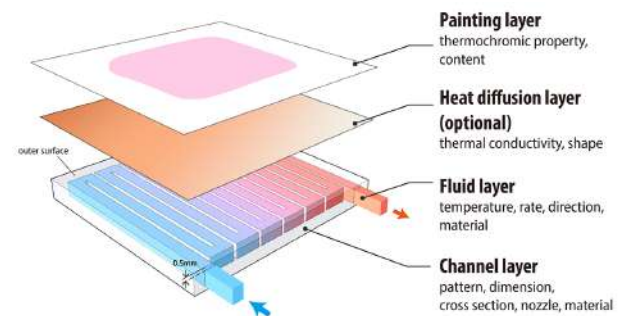


Figure 2: The basic four-layered structure of Thermotion composites. The *channel layer* is the container of the thermofluids with fluidic channels embedded under the object surface. The *fluid layer* refers to the thermofluids with different temperatures working as cold and heat sources. The *heat diffusion layer* is an optional layer used to intensify the temperature diffusion along the outer surface. The *painting layer* contains thermochromic paints with customized contents to react to temperature changes.

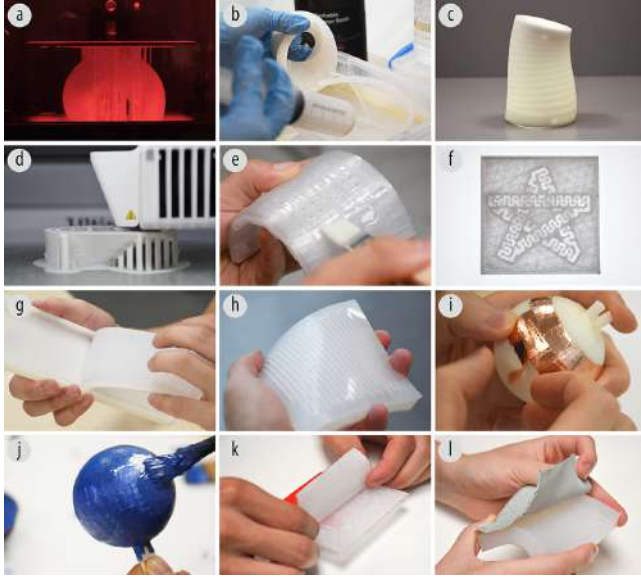


Figure 3: Fabrication pipeline (step1-3) of Thermotion composites.

2.2 Fabrication Pipeline

2.2.1 Step1: Fabricate Channel Layer. We tested stereolithography (SLA) (Figure 3a-c) and fused deposition modeling (FDM) (Figure 3d-f) techniques to fabricate a rigid channel layer, while both need additional post-processing steps, i.e., draining the inner residual resin for SLA prints with syringes (Figure 3b) and sealing the curved outer surface of the fluidic channels for FDM prints with UV glue (Figure 3e). We used 3D printers of ELEGOO Mars 2 Pro, FormLabs Form 2, and Ultimaker S5, with materials of water-washable resin, clear resin, PC, ABS, and PETG. We also tested silicone casting method to produce a flexible channel layer with 3D-printed molds (Figure 3g) and post-sealing procedure (Figure 3h).

2.2.2 Step2: Fabricate Heat Diffusion Layer. In this project, the heat diffusion layer was generally fabricated by cutting copper tapes into designed shapes and pasting them at the corresponding position on the outer surface of the channel layer (Figure 3i).

2.2.3 Step3: Fabricate Painting Layer. The easiest way we tested to fabricate the painting layer is to paint the thermochromic paints directly on the object's outer surface (Figure 3j). We also tested by first painting the thermochromic paints on a piece of everyday sheet-like material (e.g., paper or fabric) or using printed thermochromic papers and then attaching them to the object surfaces as the painting layer (Figure 3k,l) in order to conserve the familiarity and aesthetics of the surface material.

2.2.4 Step4: Generate Fluid Layer. To generate the fluid layer, we implemented a low-cost hydraulic circuit, which could programmably control the flow rate, temperature, and direction and then activate the programmed animation effects (Figure 4). Two flows of water are drawn by two peristaltic pumps (Kamoer KHS-SW3N40) from the hot and the cold tanks, respectively, and then merged to generate the expected thermofluidic flow with adjustable

flow rate and temperature. The flow direction is switched by an "H-bridge" module, which works similarly to the H-bridge circuit but uses four solenoid valves (ZhiRongHuaGuan Fa2021A) as the switches (Figure 4b). We used Arduino Nano and L298N driver boards to control the pump flow rates by PWM signals, as well as the on-off states of the four solenoid valves in the "H-bridge" module by bit signals.

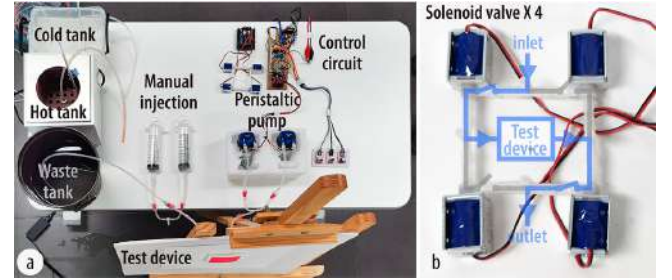


Figure 4: Electrical hydraulic circuit to generate the fluid layer. (a) Overview of the prototype, where part of the connecting tubes is omitted. (b) The H-bridge module to switches the fluidic direction and the equivalent schematics.

In addition, we also tested two non-electrical methods, including manual injection with syringes and natural force to activate the thermofluidic flow as a passive method in structures like communicating vessels.

3 ANIMATION EFFECT DESIGN

By varying the geometry pattern of the fluidic channels in the channel layer, Thermotion can create various types of animation effect primitives for one activating area to change the color from one to another (Figure 1a-e). Thermotion can also create variations of the primitives on two properties, which are *continuity* (Figure 1f) and *direction*, by varying the temperature and direction of the thermofluidic flow in the fluid layer.

Two modification effects can be designed to expand the visual expressiveness. Thermotion can create the *blur effect* of the animation by adding a highly thermally conductive material (e.g., copper tape in this work) as the heat diffusion layer to smooth the temperature distribution as well as the color distribution (Figure 1g). Thermotion can also create *multi-color effect* using thermochromic paints with multiple color-changing stages in the painting layer (Figure 1h).

To help replicating, we provided the 3D models for all examples in Figure 1¹. All above examples except Figure 1h change the color from white to red, using 45°C discoloration red paints, which are pre-heated by 60°C water to display white and then perfused with 20°C water to activate the turning-red animation effect. Figure 1h uses liquid crystal, which changes multiple colors in sequence from 28-34°C. The device is perfused with 20°C and 60°C water in turn.

¹<https://github.com/theSkai/Thermotion/tree/main/3D%20model>

4 APPLICATION SCENARIOS

We detail three application scenarios that are implied by the various animation effects of Thermotion. The 3D printing models used in the following examples are provided for replicating².

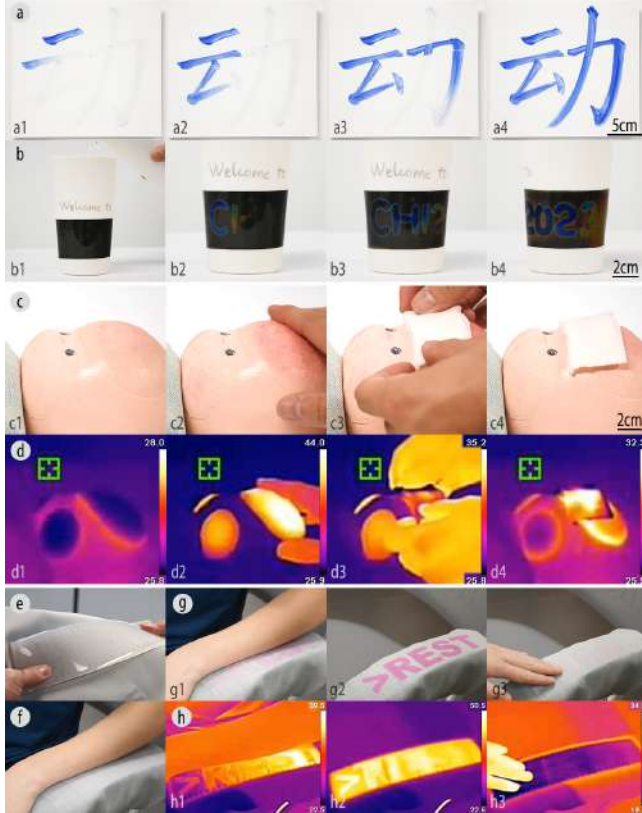


Figure 5: Application scenarios. (a,b)Color-changing artifacts. (c,d)Interactive teaching aids. (e-h)Dynamic skins for ambient display, with a flexible composites(e) on a armrest(f). Photography and thermography images with the same numerical label of the same example are captured at the same moment.

4.1 Color-changing Artifacts

Thermotion can dynamically change the color of two and three dimensional objects to expand the artifact's expressiveness. Figure 5a shows a dynamic calligraphy work for the Chinese character "movement". We installed a 3D-printed flow channel on the back of the paper written with thermochromic paints. The static Chinese characters on the paper can dynamically show the writing process, making the works more aesthetically expressive. Figure 5b shows a dynamic water cup. The inside of the water cup is embedded with a fluidic channel that communicates with the inside of the cup. When the user pours hot water into the cup, driven by gravity, the hot water automatically flows into the flow channel, activating the surface of the cup wall to appear with a writing animation.

²<https://github.com/theSkai/Thermotion/tree/main/3D%20model>

4.2 Interactive Teaching Aids

The higher dynamic display with visual and thermal feedback of Thermotion inspired us to explore the applications of multi-modal interactive teaching aids. Figure 5c,d shows an example of a dynamic physiology teaching aid, which uses a blurry, continuous, reversed-looped spread primitive to show the fever symptoms. Specifically, when we see redness on the model's forehead and cheeks, we feel warm touching these areas. When we take appropriate treatments, such as applying a cold compress, the redness will fade, and the model's temperature will return to normal. This interaction agrees with everyday human habits of handling a fever.

4.3 Dynamic Skins for Ambient Display

Thermotion can provide non-intrusive visual and haptic feedback at the same time on an object's surface as dynamic skins for ambient display. Figure 5e-h show flexible composites made with silicone and attached to a fabric of an armrest. After people sit for a long time, the armrest will present a visual spaced-wiping effect (Figure 5g1), and at the same time, it will be accompanied by an uneven moving temperature distribution presenting a dynamic alternating warm and cold thermal feedback (Figure 5h1). Both thermal and visual feedback will remind the user to get up and rest in a non-intrusive way. When the user gets up, the temperature of the armrest is designed to remain high enough (Figure 5h2) so that the letters on the armrest will not disappear (Figure 5g2), prompting the user not to return to the seat. When the rest time is over, the user will see the text disappear (Figure 5g3), and the armrest will return to normal temperature (Figure 5h3), reminding the user to return to the seat.

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